Saithan Panchaharan

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Education

University of Toronto Sep. 2024 – June 2029 *(Expected)*
BSc in Computer Science (Co-Op) *Toronto, ON*

- **Coursework**: *Shell Programming, Data Structures & Algorithms, Microprogramming, Assembly Programming, Operating Systems, OOP, Statistics & Data Analysis*

Experience

Bellwoods Strategic Capital Sep. 2026 – Present
Full-Stack Developer *Toronto, ON*

- Implementing features & fixing bugs in **Thrive: Happy Habits**, a **React Native**, **Expo** and **Django REST Framework** app (∼90 API routes, 16 Jest suites)
- Scoped **5** MVP user stories with the client in team meetings, prototyping user flows in **Figma**, & rebuilding the quest filtering logic
- Delivering biweekly **Android APK** builds for client review, targeting **Google Play** and **iOS App Store** releases

Projects

Sportfolio · *Live Demo* August 2026

- Built a **Spring Boot** API that models professional athletes as tradable stocks, with a pricing engine that converts real MLB performance data into daily price movements
- Designed a **PostgreSQL** database to support trading, portfolio valuation, and historical price tracking across hundreds of players
- Integrated two sports-data APIs to ingest live stats and season projections, using a caching layer to recalculate prices at no additional cost without re-fetching on every run
- Deployed the API on **Railway** and resolved production issues (server crashes from large-scale recalculations, data errors from mid-season trades) through log and API-level debugging

LineTracker · *Live Demo* July 2026

- Built a full-stack real-time alert platform that monitors live sports betting odds and stock prices, notifying users via email when user-defined thresholds are crossed
- Engineered a **Flask REST API** with **APScheduler** polling live data every minute across **SharpAPI** and **Yahoo Finance**, with in-memory caching to stay within rate limits and a **Supabase** backend for persistent alert storage
- Designed a multi-step betting interface that fetches, groups, and deduplicates live odds across multiple sportsbooks in real time, normalizing inconsistent team names and logos across data sources
- Rebuilt as a native iOS app in **SwiftUI** with **Swift Concurrency**, using a **URLSession**-based networking layer mirroring the existing REST API and **Keychain**-backed session storage

YouTube Video Summarizer · *Live Demo* May 2026 – June 2026

- Built a full-stack web app using **React** & **Flask** that fetches YouTube transcripts and generates summaries using **Claude API**
- Integrated **Supadata API** to fetch transcripts, bypassing cloud IP restrictions on server-side requests
- Deployed frontend on **Vercel** & backend on **Render** with environment-based API key management and CORS configuration for cross-origin communication
- Incorporated **Web Speech API** for text-to-speech playback & implemented transcript caching to eliminate redundant API calls when switching expertise levels

Technical Skills

Languages & Frameworks: Python, Java, JavaScript, C/C++, SQL, React Native, React, Django, Spring Boot, Flask

Tools: PostgreSQL, Docker, Git/GitHub, GitHub Actions, Jest, Figma, Railway, Vercel, Render, Linux